

Parameter-Efficient Multilinear Readout for Quantum Reservoir Computing

Van Tien Nguyen

Department of Electrical Engineering
The University of Texas at San Antonio
San Antonio, TX, USA
tien.nguyen@utsa.edu

Panagiotis Markopoulos

Departments of Computer Engineering and Computer Science
The University of Texas at San Antonio
San Antonio, TX, USA
panagiotis.markopoulos@utsa.edu

Abstract—Quantum Reservoir Computing (QRC) uses fixed quantum dynamics as a high-dimensional feature map. Still, standard pipelines flatten reservoir measurements and train a linear readout, ignoring the natural tensor structure across time steps, qubits, and observables. We propose a multilinear readout for quantum reservoir computing (ML-QRC), a tensor-structured neural readout that compresses $\mathcal{X} \in \mathbb{R}^{T \times n \times F}$ through mode-wise projections, $\mathcal{Y} = \mathcal{X} \times_1 \mathbf{U}_T \times_2 \mathbf{U}_n \times_3 \mathbf{U}_F$. Relative to a dense hidden-width-64 neural readout, this reduces the feature-to-hidden parameter count from $64TnF$ to $d(T+n+F) + 64d^3$, giving a $12.1\times$ feature-block compression at $T = 50$, $n = 6$, $d = 8$, and $F = n + \binom{n}{2}$. Across Mackey-Glass forecasting and TFIM phase classification, ML-QRC remains statistically competitive with Ridge on Mackey-Glass and obtains the numerically highest full-data TFIM accuracy among tested methods, while using $12.0\times$ fewer parameters than the full dense MLP. Its clearest benefit is improved low-data stability relative to neural MLP readouts on Mackey-Glass; this advantage does not transfer to TFIM, where Ridge performs best at 50% data. Overall, ML-QRC is best viewed as a tensor-structured neural readout that compresses dense MLP readouts while preserving QRC tensor structure, rather than as a replacement for simple linear/Ridge readouts in the tested regime.

Index Terms—quantum reservoir computing, multilinear transformation, efficient quantum machine learning, quantum phase classification, time-series forecasting

I. INTRODUCTION

Reservoir computing (RC) is a recurrent machine-learning paradigm in which a fixed dynamical system maps inputs into a rich feature space while only a lightweight readout layer is trained [1], [2]. Quantum Reservoir Computing (QRC) replaces the classical reservoir with a quantum many-body system, exploiting superposition, entanglement, and quantum chaos to generate exponentially rich feature representations from modest qubit counts [3]–[6]. Because the reservoir is fixed, QRC sidesteps the barren-plateau problem of variational quantum circuits [7], making it attractive for NISQ hardware [8]. Hardware demonstrations have now matured across superconducting processors [9], photonic platforms [10], and a 108-qubit Rydberg array—the largest quantum machine-learning demonstration to date [11].

Despite this progress, the *readout layer* has received almost no architectural attention. Most standard QRC pipelines collect expectation values $\langle O_f \rangle_t = \text{Tr}[\rho(t) O_f]$ for F observables at T time steps, concatenate them into a flat vector $\mathbf{x} \in \mathbb{R}^{TnF}$,

and fit a linear map $\hat{y} = \mathbf{w}^\top \mathbf{x}$. This incurs three costs: (i) collapsing time, qubit, and observable axes into one unstructured dimension discards multi-way correlations; (ii) a dense readout requires $C \cdot TnF$ parameters, growing as $O(n^4)$ when $T = O(n)$ and $F = O(n^2)$; and (iii) treating all (t, i, f) triplets symmetrically ignores the heterogeneous roles of temporal dynamics, qubit-qubit correlations, and observable-basis diversity.

Tucker-factored readout heads [12], [13] are well established in classical deep learning for compressing multi-way feature tensors from convolutional networks. This principle was later extended to hybrid quantum–classical architectures through quantum-based tensor contraction layers [14]. Motivated by these classical and hybrid precedents, applying mode-wise contraction to QRC raises a distinct design question: can projections along the time, qubit, and observable modes compress quantum-reservoir measurements without sacrificing predictive signal?

Contributions. We propose a multilinear transformation for quantum reservoir computing (ML-QRC), a parameter-efficient readout that applies learned mode-wise projections directly to the measurement tensor:

$$\mathcal{Y} = \mathcal{X} \times_1 \mathbf{U}_T \times_2 \mathbf{U}_n \times_3 \mathbf{U}_F.$$

By the parameter formula, the feature-block compression relative to a dense hidden-width-64 neural readout increases from $3.85\times$ to $52.24\times$ as n grows from 4 to 10 under $T = 50$, $d = 8$, and $F = n + \binom{n}{2}$. We evaluate ML-QRC on Mackey-Glass forecasting and TFIM phase classification using an *identical* hidden-width-64 readout head and a size-matched MLP-S control in both tasks; as summarized in the abstract, ML-QRC-M is statistically competitive with Ridge on Mackey-Glass, achieves the numerically highest full-data TFIM accuracy, and uses $12.0\times$ fewer parameters than the dense MLP at $n = 6$, while its low-data stability advantage over neural readouts does not transfer to TFIM. These results position ML-QRC as a compressor of dense MLP readouts that preserves QRC’s time–qubit–observable structure, not as a lower-parameter replacement for simple Linear/Ridge readouts.

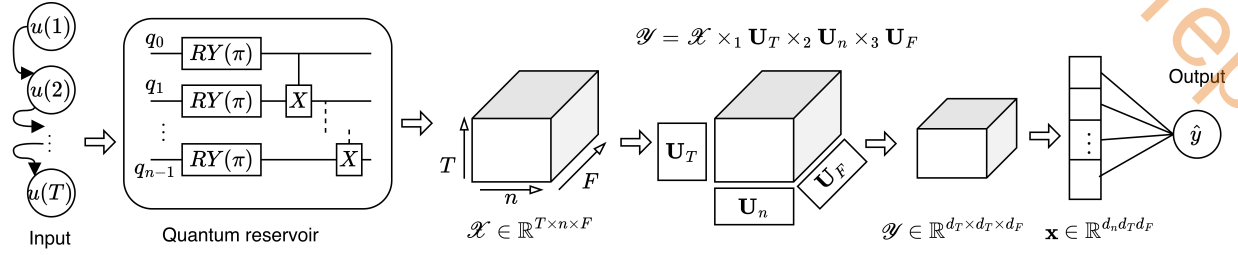


Fig. 1. The proposed multilinear transformation for QRC. Input sequences are encoded into an n -qubit reservoir producing a feature tensor $\mathcal{X} \in \mathbb{R}^{T \times n \times F}$. The projection compresses each mode independently via $\mathbf{U}_T$, $\mathbf{U}_n$, $\mathbf{U}_F$ before a neural readout head produces $\hat{\mathbf{y}}$.

II. BACKGROUND AND RELATED WORK

Quantum Reservoir Computing. Fujii and Nakajima [3] established QRC universality; subsequent work boosted memory capacity [4], studied quantum reservoir processing [5], dynamical phase transitions [6], and echo-state conditions under noise [9].

Classical RC. Echo State Networks [1] and Liquid State Machines [2] established RC as a practical paradigm. Next-Generation RC [15] achieves near-machine-precision Mackey-Glass forecasting (NMSE $\approx 10^{-4}$) without a physical reservoir, setting the classical ceiling.

Tensor contraction layers. Kossaifi et al. [12], [13] showed that fully connected CNN readouts can be replaced by mode-wise tensor contractions without sacrificing accuracy. Nguyen and Markopoulos [14] extended this principle to a hybrid quantum–classical setting: on VGG-19/CIFAR-100, QTCL achieved 71.67% accuracy with 98.68% head-parameter savings, outperforming TCL by 4.19 percentage points at comparable compression, although it underperformed TCL under aggressive AlexNet compression.

III. MULTILINEAR QUANTUM RESERVOIR COMPUTING

A. Quantum Reservoir Feature Tensor

QRC operates sequentially over T time steps. At each step t , input $u(t)$ is encoded into an n -qubit state evolving under a fixed reservoir unitary U_{res} ; then F Pauli observables are measured:

$$\mathcal{X}[t, i, f] = \text{Tr}[\rho(t) O_f^{(i)}], \quad t \in [T], \quad i \in [n], \quad f \in [F]. \quad (1)$$

The *time axis* encodes causal dynamics and fading memory; the *qubit axis* encodes entanglement-induced correlations including $\langle Z_i \rangle$ and $\langle Z_i Z_j \rangle$; the *observable axis* spans different Pauli bases. The standard flat readout collapses all three axes:

$$\hat{\mathbf{y}} = \mathbf{W}_{\text{out}} \text{vec}(\mathcal{X}) \in \mathbb{R}^C, \quad (2)$$

where $\mathbf{W}_{\text{out}} \in \mathbb{R}^{C \times TnF}$ requires $C \cdot TnF$ parameters.

B. Multilinear Projection Readout

We replace (2) with a multilinear projection layer parametrized by factor matrices $\mathbf{U}_T \in \mathbb{R}^{d_T \times T}$, $\mathbf{U}_n \in \mathbb{R}^{d_n \times n}$, $\mathbf{U}_F \in \mathbb{R}^{d_F \times F}$:

$$\mathcal{Y} = \mathcal{X} \times_1 \mathbf{U}_T \times_2 \mathbf{U}_n \times_3 \mathbf{U}_F, \quad (3)$$

where each entry of $\mathcal{Y}$ is

$$\mathcal{Y}[j_T, j_n, j_F] = \sum_{t, i, f} \mathcal{X}[t, i, f] U_T[j_T, t] U_n[j_n, i] U_F[j_F, f].$$

$\mathcal{Y}$ is flattened and mapped to an H -dimensional hidden representation:

$$\mathbf{h} = \sigma(\mathbf{W}_h \text{vec}(\mathcal{Y}) + \mathbf{b}_h) \in \mathbb{R}^H, \quad (4)$$

where $\mathbf{W}_h \in \mathbb{R}^{H \times d_T d_n d_F}$ and σ is ReLU. The task output is

$$\hat{\mathbf{y}} = \mathbf{W}_o \mathbf{h} + \mathbf{b}_o \in \mathbb{R}^C, \quad (5)$$

with $\mathbf{W}_o \in \mathbb{R}^{C \times H}$.

Setting $d_T = d_n = d_F = d$, the feature-to-hidden parameter count falls from $HTnF$ for the dense MLP to $d(T+n+F) + Hd^3$ for ML-QRC. With $H = 64$, $n = 6$, $T = 50$, $F = 21$, and $d = 8$, this reduces the feature block from 403,200 to 33,384 parameters, a $12.1\times$ reduction. Both evaluated tasks use a single output, $C = 1$.

C. Training

All parameters $\Theta = \{\mathbf{U}_T, \mathbf{U}_n, \mathbf{U}_F, \mathbf{W}_h, \mathbf{b}_h, \mathbf{W}_o, \mathbf{b}_o\}$ are trained jointly by backpropagation through the three-mode products, so each factor learns a mode-specific projection. Stage 1 compresses $\mathcal{X}$ via three mode products in (3) at cost $O(d_T T n F + d_n d_T n F + d_F d_T d_n F)$, dominated by the first contraction when $d_k \ll T, n, F$. Stage 2 applies the neural readout head in Eqs. (4)–(5).

IV. NUMERICAL STUDIES

A. Experimental Setup

Reservoir. All experiments use a hardware-efficient strongly entangling ansatz [16] implemented in PennyLane [17] with PyTorch, with $T = 50$, $n = 6$, $F = n + \binom{n}{2} = 21$ observables ($\{Z_i, Z_i Z_j\}$), and flat dimension $T \times n \times F = 6,300$.

Noise model. We apply an analytical depolarizing channel ($p = 0.01$): expectation values are scaled by $(1 - 4p/3)$ for $\langle Z_i \rangle$ and $(1 - 4p/3)^2$ for $\langle Z_i Z_j \rangle$, exact for independent single-qubit channels.

Readout baselines. All neural readouts below use ReLU hidden activations and a linear output head (no final activation). *Linear*: ordinary least squares on the flattened reservoir feature vector. *Ridge*: ℓ_2 -regularized readout, with α tuned by cross-validation. *MLP*: a one-hidden-layer network applied to $\text{vec}(\mathcal{X})$, using a $TnF \rightarrow 64$ fully connected layer. *MLP-S*:

TABLE I

MAIN COMPARISON: MACKEY-GLASS (NMSE $\downarrow$, $n = 6$) AND TFIM PHASE CLASSIFICATION (ACCURACY % $\uparrow$, $n = 6$); PARAMETER COUNTS SHARED ACROSS TASKS. *MLP-S IS SIZE-MATCHED TO ML-QRC-M. BOLD = BEST VALUE PER COLUMN AMONG QRC READOUTS (ESN/NG-RC USE A NON-QUANTUM RESERVOIR, SHOWN ONLY AS A CLASSICAL CEILING, EXCLUDED FROM BOLDING).

Method	Params	MG full	MG 10%	TFIM full	TFIM 50%
ESN [1]	—	0.0002	0.0002	—	—
NG-RC [15]	—	<0.0001	<0.0001	—	—
Linear	6,301	0.0218 $\pm$ 0.0000	0.6652$\pm$0.0000	64.8 $\pm$ 0.0	53.6 $\pm$ 2.1
Ridge	6,301	0.0085 $\pm$ 0.0000	0.7572 $\pm$ 0.0000	69.9 $\pm$ 0.0	72.6$\pm$1.0
MLP	403,329	0.0060 $\pm$ 0.0012	1.6105 $\pm$ 0.2529	69.5 $\pm$ 0.9	67.4 $\pm$ 1.4
MLP-S*	33,537	0.0058$\pm$0.0011	1.9463 $\pm$ 0.4839	70.4 $\pm$ 0.6	68.6 $\pm$ 1.5
Ridge-PCA- d^3	513	0.0085 $\pm$ 0.0000	0.7572 $\pm$ 0.0000	69.6 $\pm$ 0.0	72.6$\pm$0.5
Tensor-Ridge	385	0.0265 $\pm$ 0.0000	0.8258 $\pm$ 0.0000	67.0 $\pm$ 0.0	62.1 $\pm$ 2.1
ML-QRC (ours)	33,513	0.0086 $\pm$ 0.0013	0.9684 $\pm$ 0.1581	70.5$\pm$1.5	66.6 $\pm$ 1.8

PCA-512 followed by a one-hidden-layer MLP with 64 hidden units, explicitly *size-matched* to ML-QRC-M ($\approx 33,537$ parameters, vs. ML-QRC-M’s 33,513) in both tasks. *ML-QRC-M*: the proposed multilinear readout with $d = 8$, which first maps $\mathcal{X} \in \mathbb{R}^{T \times n \times F}$ to $\mathcal{Y} \in \mathbb{R}^{8 \times 8 \times 8}$ via mode-wise projections, then a $512 \rightarrow 64$ fully connected layer, for 33,513 total parameters versus 403,329 for the full flat MLP (a $12.0 \times$ network-level reduction; see Sec. III). All parameter counts are shared across Mackey-Glass and TFIM (single output unit). We apply this same hidden-width-64 head across $d \in \{4, 8, 16, 32\}$ and both tasks, giving 4,533/33,513/263,505/2,099,745 parameters, respectively (Sec. IV-C). *Ridge-PCA- d^3* : PCA to $\min(d^3, n_{\text{train}} - 1, TnF) = 512$ dimensions, followed by Ridge (513 parameters). *Tensor-Ridge*: HOSVD Tucker compression followed by Ridge at effective ranks (8, 6, 8) (385 parameters).

We use Adam (lr 10^{-3} , weight decay 10^{-5} , 300 epochs, batch size 32) and report results over 10 random seeds.

B. Main Comparison: Mackey-Glass and TFIM

Mackey-Glass. We use the standard chaotic benchmark ($\beta = 0.2$, $\gamma = 0.1$, $q = 10$, $\tau = 17$) [1], [15] with 3,000 training / 1,000 test samples, plus a 10%-data regime (300 samples); performance is NMSE ($\downarrow$). **TFIM.** We classify ground states of the transverse-field Ising model with $n = 6$, $g/J \in [0.2, 1.8]$ (8 ferromagnetic, 8 paramagnetic couplings), 200 Z -basis sequences per coupling, 3,200 samples (70/30 split), plus a 50%-training regime; performance is top-1 accuracy ($\uparrow$).

Results (Table I). On Mackey-Glass, all QRC readouts trail ESN/NG-RC by roughly an order of magnitude, reflecting the 6-qubit reservoir’s limited memory [3] rather than readout design. Ridge attains the best full-data NMSE (0.0085); ML-QRC-M is statistically indistinguishable (0.0086 ± 0.0013). MLP-S edges out ML-QRC-M on full data (0.0058) but collapses under scarcity (1.9463, worse than the full MLP’s 1.6105), while ML-QRC-M is far more stable among *neural* readouts (0.9684)—though Linear (0.6652) and Ridge (0.7572) remain the best low-data performers overall, so this advantage is relative to other neural readouts only. On TFIM, ML-QRC-M’s full-data accuracy ($70.5 \pm 1.5\%$) is numerically highest but inside the overlapping error bars of MLP-S (70.4%) and MLP (69.5%)—a tie we now reflect by

holding the true best entry per column rather than only our own method. This edge does not transfer to TFIM at 50% data, where Ridge and Ridge-PCA- d^3 (both 72.6%) beat every neural readout, including ML-QRC-M (66.6%). Ridge’s TFIM accuracy is oddly higher at 50% than 100% data; since α is re-tuned per regime from the same pool under different splits, we read this as split-dependent rather than a genuine low-data benefit, and flag it as an unverified robustness question. Both tasks improve substantially over static-HOSVD Tensor-Ridge (Mackey-Glass: $0.0265 \rightarrow 0.0086$; TFIM: $67.0 \rightarrow 70.5\%$), confirming *learned* Tucker factors add value beyond a fixed-rank projection. ML-QRC-M uses $5.3 \times$ more parameters than plain Linear/Ridge; we position it as a compressor of dense MLP readouts, not a parameter-count win over Linear/Ridge.

C. Scaling and Projection-Dimension Studies

Qubit scaling (Table IIa). Because the readout head is fixed at hidden width 64 for every d and n , ML-QRC-M’s parameter count is nearly n -independent (33,409–33,817) while the raw flat dimension grows from 2,000 to 27,500; the flat-to-ML-QRC ratio rises only from $0.06 \times$ to $0.81 \times$ in *both* tasks—ML-QRC-M uses more parameters than a bias-free linear readout throughout, with compression recovered only relative to the dense MLP ($O(64 TnF)$). MLP-S scales better with n in both tasks: on Mackey-Glass the gap widens from $n = 6$ onward (0.0058 vs. 0.0086; 0.0097 vs. 0.0136 at $n = 10$); on TFIM, MLP-S reaches 78.4% vs. ML-QRC-M’s 74.1% at $n = 10$. Only at $n = 4$, only on Mackey-Glass, does ML-QRC-M edge out MLP-S (0.0076 vs. 0.0082); on TFIM, MLP-S leads at every n . Fixing head width while n grows is precisely what erodes ML-QRC-M’s standing; scaling d with n is left for future work.

Projection dimension (Table IIb). Full-data accuracy improves monotonically with d on *both* tasks (Mackey-Glass: $0.0125 \rightarrow 0.0067$; TFIM: $68.7 \rightarrow 71.4\%$), revising our earlier, architecture-confounded hypothesis that rank helps only temporally structured inputs. The cost is steep: parameters grow as d^3 , so ML-QRC-XL (2,099,745) is $5.2 \times$ larger than the dense MLP; only S/M/L stay smaller ($89 \times / 12.0 \times / 1.5 \times$). Since d and parameter count grow together, we cannot separate a rank effect from a size effect; holding parameters fixed while varying d is left for future work. At reduced

TABLE II
SCALING STUDIES ON MACKEY-GLASS (NMSE ↓) AND TFIM (ACCURACY % ↑); PARAMETER COUNTS SHARED ACROSS TASKS (SEC. IV-A).
BOLD = BEST VALUE PER COLUMN. (A) QUBIT SCALING AT FIXED $d = 8$ ([†]FROM TABLE I; RATIO = FLAT DIM. / ML-QRC PARAMS). (B)
PROJECTION-DIMENSION SWEEP AT FIXED $n = 6$.

(a) Qubit scaling, fixed $d = 8$								
n	Flat	ML-QRC	Ratio	MG MLP-S	MG ML-QRC	TFIM MLP-S	TFIM ML-QRC	
4	2,000	33,409	0.06×	0.0082±0.0019	0.0076±0.0019	68.0±0.9	65.7±0.7	
6 [†]	6,300	33,513	0.19×	0.0058±0.0011	0.0086±0.0013	70.4±0.6	70.5±1.5	
8	14,400	33,649	0.43×	0.0096±0.0025	0.0132±0.0022	77.3±0.4	72.9±0.8	
10	27,500	33,817	0.81×	0.0097±0.0020	0.0136±0.0021	78.4±0.4	74.1±1.6	

(b) Projection-dimension sweep, fixed $n = 6$						
Config	Params	d	MG full	MG 10%	TFIM full	TFIM 50%
MLP (ref.)	403,329	—	0.0060±0.0012	1.6105±0.2529	69.5±0.9	67.4±1.4
MLP-S (ref.)	33,537	—	0.0058±0.0011	1.9463±0.4839	70.4±0.6	68.6±1.5
ML-QRC-S	4,533	4	0.0125±0.0021	0.9336±0.3099	68.7±1.0	65.4±1.9
ML-QRC-M	33,513	8	0.0086±0.0013	0.9684±0.1581	70.5±1.5	66.6±1.8
ML-QRC-L	263,505	16	0.0075±0.0014	0.9206±0.1452	70.8±1.0	68.2±1.6
ML-QRC-XL	2,099,745	32	0.0067±0.0007	0.9864±0.1063	71.4±1.0	69.2±0.8

data the tasks diverge: every ML-QRC variant on Mackey-Glass (NMSE 0.92–0.99) stays far more stable than MLP-S (1.9463) or MLP (1.6105); on TFIM, accuracy again rises with d (65.4%→69.2%) but never surpasses Ridge’s 72.6%, so the Tucker readout offers no low-data edge over a simple regularized linear baseline. Separately, both MLP and MLP-S exceed an NMSE of 1 at 10% Mackey-Glass data (1.6105, 1.9463)—worse than predicting the training mean; we have not separately tuned early stopping for the MLP baselines in this regime, so a better-tuned MLP could narrow, though likely not close, this stability gap, which we flag as a limitation rather than adjust post hoc.

V. CONCLUSION

We introduced ML-QRC, a Tucker-factored multilinear neural readout that preserves the time–qubit–observable structure of QRC measurements. Compared with a dense hidden-width-64 MLP readout, ML-QRC-M reduces network size by $12.0\times$ at $n = 6$ while remaining competitive on Mackey-Glass and achieving the numerically highest full-data TFIM accuracy, though within the margin of statistical variation. Future work includes lower-scaling tensor heads, larger reservoirs, finite-shot noise, parameter-matched rank sweeps, and hardware validation.

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